

A Gate-Structured Observational and Falsification Program for the Integrated Toroidal-Syntropic Model

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I. INTRODUCTION

A central challenge in modified gravity and dark fluid theories is achieving multi-scale consistency without ad-hoc parameter tuning [1–3]. In the Integrated Toroidal-Syntropic Model (ITSM), the observable vacuum is described as an active finite-density Bose-Einstein condensate with compact toroidal (T^3) topology, open-system syntropic exchange, and emergent infrared acceleration dynamics.

Companion papers establish present-epoch scale matching and claim hygiene (Paper I) [4] and anisotropic Casimir vacuum stress on rectangular T^3 lattices (Paper II) [5]. Furthermore, the exact conformal Weyl trace fixes the matter coupling $C_m \equiv 1.0$, while Baryonic Tully-Fisher scale matching determines the condensate decay constant $f = 1/\sqrt{4\pi G}$, yielding the exact low-energy AQUAL action with strength $\alpha \equiv 4\pi G(C_m/f)^2 = 1.0$ [6].

In this paper (Paper III), we present the gate-structured observational program and falsification framework that directly confronts this first-principles action with empirical tests across three distinct regimes:

1. **Acoustic Phonon Resonances (Gate VOR-001):** Discrete Bogoliubov-de Gennes modes on compact T^3 .
2. **Solar System Screening (Gate SCR-001):** Landau critical velocity disruption against Cassini radar bounds.
3. **Relativistic Lensing (Gate LEN-001):** Null geodesics and shear profiles in the scale-compensator metric.

II. ACOUSTIC RESONANCES ON COMPACT T^3

The complex condensate order parameter is $\Phi = \frac{\rho}{\sqrt{2}}e^{i\Theta}$, where topological winding sectors are labelled by integers $\mathbf{n} = (n_1, n_2, n_3) \in \mathbb{Z}^3$ around the non-contractible 1-cycles of $T^3 = S^1 \times S^1 \times S^1$.

Linearizing perturbations $\Phi = (\rho_0 + \delta\rho)e^{i(\Theta_0 + \vartheta)}$ about a steady circulation background $\mathbf{v}_0 = \frac{\hbar}{m}\nabla\Theta_0 = \frac{\hbar}{m}\left(\frac{2\pi n_1}{L_1}, \frac{2\pi n_2}{L_2}, \frac{2\pi n_3}{L_3}\right)$ yields the Bogoliubov-de Gennes dispersion relation:

$$\omega_{\mathbf{k}} = \mathbf{v}_0 \cdot \mathbf{k} + \sqrt{c_s^2 k^2 + \left(\frac{\hbar k^2}{2m}\right)^2}, \quad (1)$$

where $c_s = c/\sqrt{3}$ is the relativistic conformal sound speed, and the allowed discrete wavevectors are quantized by the compact box lengths:

$$\mathbf{k} = 2\pi \left(\frac{m_1}{L_1}, \frac{m_2}{L_2}, \frac{m_3}{L_3}\right), \quad \mathbf{m} \in \mathbb{Z}^3 \setminus \{(0, 0, 0)\}. \quad (2)$$

For an interstellar/cluster cavity of characteristic scale $L \approx 3.0$ pc (9.26×10^{16} m), the fundamental acoustic mode frequencies $f_{\mathbf{m}} = \omega_{\mathbf{m}}/(2\pi)$ evaluate to:

$$f_{(0,0,1)} = 1.882 \text{ nHz}, \quad f_{(-1,0,0)} = 1.449 \text{ nHz}, \quad (3)$$

with Doppler splitting $\Delta f \approx 0.433$ nHz induced by the topological superflow. This natural discrete spectrum matches the 1–3 nHz stochastic acoustic band detected by pulsar timing array collaborations (NANOGrav 15-year data set) [7].

III. LANDAU CONDENSATE DISRUPTION SCREENING

A fatal objection to unshielded scalar-tensor theories is the generation of fifth forces in the Solar System. In ITSM, screening arises dynamically from microscopic condensate physics rather than fine-tuned chameleon potentials.

The matter interaction Lagrangian is $\mathcal{L}_{\text{int}} = -C_m \rho_b \psi$. When the induced scalar gradient $|\nabla\psi|$ generates an effective superfluid velocity $v_s = \frac{\hbar}{m}|\nabla\psi|$ exceeding the Landau

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TABLE I. Quantitative Falsification Criteria for ITSM.

Observable	Threshold / Band	Falsifier Condition
PTA Acoustic Mode	1.0–3.5 nHz	Zero spectral power
Solar PPN $\Delta\gamma$	$< 2.3 \times 10^{-5}$	Measured $\Delta\gamma > 10^{-4}$
Lensing / Dyn Ratio	$M_{\text{lens}}/M_{\text{dyn}} \equiv 1.0$	Measured ratio $\neq 1.0 \pm 0.1$
SPARC Global χ^2_ν	$\chi^2_\nu \leq 7.5$	MCMC fit $\chi^2_\nu > 15.0$

critical velocity $v_{\text{crit}} = c_s$, macroscopic phase coherence is disrupted. Dynamic vortex loops nucleate, quenching the coherent scalar field gradient.

The screened fifth-force acceleration is saturated at $a_5^{\text{screen}} \leq a_0$. The post-Newtonian parameter deviation $\Delta\gamma = |\gamma_{\text{PPN}} - 1|$ is given by:

$$\Delta\gamma(r) = 2C_m^2 \left(\frac{a_5^{\text{screen}}(r)}{g_N(r)} \right) = 2 \left(\frac{a_0 r^2}{GM_\odot} \right). \quad (4)$$

Evaluating at key Solar System baselines yields:

$$\Delta\gamma(R_\odot) \approx 8.77 \times 10^{-13}, \quad (5)$$

$$\Delta\gamma(1 \text{ AU}) \approx 4.05 \times 10^{-8}. \quad (6)$$

The predicted deviation is over 500 times smaller than the observational upper limit established by the Cassini spacecraft radar time-delay experiment ($|\gamma_{\text{PPN}} - 1| \leq 2.3 \times 10^{-5}$) [8], guaranteeing strict Solar System compliance.

IV. RELATIVISTIC GRAVITATIONAL LENSING

Photons couple to the physical conformally rescaled metric $\tilde{g}_{\mu\nu} = e^{2\psi} g_{\mu\nu}^{\text{GR}}$. Because the Maxwell action is conformally invariant in 4 spacetime dimensions, the scalar field ψ decouples from electromagnetic rays at tree level, and null geodesics are governed by the metric $g_{\mu\nu}^{\text{GR}}$ sourced by the total stress-energy $T_{\mu\nu}^{\text{matter}} + T_{\mu\nu}^{\text{compensator}}$.

The light deflection angle $\hat{\alpha}(b)$ at impact parameter $b = R$ is:

$$\hat{\alpha}(R) = \frac{4GM_{\text{dyn}}(R)}{c^2 R}, \quad (7)$$

where $M_{\text{dyn}}(R) = R^2 g_{\text{tot}}(R)/G$ is the dynamical mass deduced from SPARC rotation curves. In the low-acceleration regime ($g_N \ll a_0$), $g_{\text{tot}} = \sqrt{a_0 g_N}$, producing a lensing boost factor:

$$\frac{\hat{\alpha}(R)}{\hat{\alpha}_{\text{GR}}(R)} = \frac{M_{\text{dyn}}(R)}{M_{\text{baryon}}(R)} = \sqrt{\frac{a_0}{g_N(R)}} > 1. \quad (8)$$

Consequently, the lensing-to-kinematic mass ratio satisfies $M_{\text{lens}}(R)/M_{\text{dyn}}(R) \equiv 1.00$ exactly, resolving galaxy cluster and galactic lensing profiles without dark matter particles.

V. FALSIFICATION TABLE

Table I summarizes the key quantitative criteria that would immediately falsify the ITSM framework.

VI. CONCLUSIONS

We have established a gate-structured observational and falsification framework for the Integrated Toroidal-Syntropic Model. By deriving the Bogoliubov acoustic resonance window (1.45–1.88 nHz), demonstrating dynamic Landau screening against Cassini bounds ($\Delta\gamma = 4.05 \times 10^{-8}$), and proving lensing-kinematic mass equivalence, ITSM provides sharp, multi-scale empirical falsifiers across astrophysical regimes.

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